

B. Ajay

Tamil Nadu, India

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🌐 ajayportfolio.in

Motivated Computer Science undergraduate with hands-on experience in Artificial Intelligence, backend development, and systems engineering. Proven ability to architect and build end-to-end applications ranging from custom C++ rendering engines to ML-driven web services and AI-powered platforms. Seeking to leverage a strong foundation in Linux environments, software optimization, and machine learning technologies in a challenging AI/ML or backend engineering role.

Education

Aalim Muhammad Salegh College of Engineering
B.E. Computer Science and Engineering

Avadi, Tamil Nadu
Expected Graduation: 2029

Technical Skills

- **Programming Languages:** Python, C/C++, JavaScript, Dart
- **Frameworks & Libraries:** Flask, React, Flutter, FastAPI, SDL2
- **Core Technologies:** Machine Learning, Computer Vision, LLM Integration (Google Gemini), NLP, Firebase, Rigid Body Physics Simulation
- **DevOps & Tools:** Docker, Git, Ngrok, SSH, Google Cloud Run
- **Systems & Databases:** Linux (Ubuntu/Xubuntu), System Administration, Networking Basics, MySQL, SQLite, Firestore

Technical Projects

OtakuCart AI (E-commerce Platform)
React, Firebase, DALL-E

[GitHub Repository](#)

- Engineered a premium, dark-themed e-commerce platform for anime enthusiasts featuring a custom glassmorphism design system.
- Integrated a smart AI-powered assistant using NLP for natural language product filtering and personalized recommendations based on user interest tracking.
- Developed a dynamic shopping cart and secure multi-step checkout process with real-time inventory previews and pop-in animations.

Disaster Crisis Management System
Flutter, FastAPI, Google Gemini AI

[GitHub Repository](#)

- Architected an AI-driven emergency response platform for real-time incident reporting, analysis, and visualization.
- Implemented automatic severity assessment using Google Gemini AI to classify incidents and generate professional briefs.
- Developed interactive maps with severity-based color coding and intelligent jitter logic for precise incident visualization.
- Built an advanced image management system with HEIC-to-JPEG conversion and a secure admin dashboard for full CRUD operations.

Ajax Engine (Custom Game/Rendering Engine)
C++

[GitHub Repository](#)

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- Engineered a lightweight, high-performance 2D game and rendering engine from scratch focusing on core mechanics.

- Designed and optimized the core game loop architecture to ensure efficient resource management, memory safety, and stable frame rates.

Lunar Lander Simulation

GitHub Repository

C++, SDL2

- Developed a comprehensive 2D physics simulation game utilizing C++ and the SDL2 rendering library.
- Implemented complex mathematical algorithms for realistic rigid body physics, environmental gravity simulation, and precise collision detection.

Litter Detection Model

GitHub Repository

Python, ML, Computer Vision

- Designed and trained a custom Computer Vision model utilizing Machine Learning classification algorithms to accurately detect environmental litter.
- Evaluated and fine-tuned model parameters to enhance detection accuracy, contributing to automated waste management solutions.

Experience & Achievements

Independent Projects

Systems Architecture & Development

- **System Deployment:** Acquired practical, hands-on experience deploying complex AI and backend software systems directly onto Linux (Ubuntu/Xubuntu) servers and Google Cloud Run.
- **Optimization & Debugging:** Cultivated strong debugging, memory management, and performance optimization skills through rigorous testing of C++ programs and server deployments.
- **Project Portfolio:** Demonstrated proficiency in full-stack project lifecycles by independently conceptualizing, developing, and successfully launching multiple cross-domain systems (Games, ML models, Web APIs, AI Platforms).